

Data Model Specification

Civil Society Anti-Money Laundering & Financial Intelligence Framework

Version 0.1 | Normative Data Specification

Status

This document defines the canonical information model for CS-AML v0.1. It is intended to be implementation-neutral and SHALL be used as the authoritative semantic reference for database schemas, APIs, graph stores, analytical tooling, exchange formats, and audit records that claim CS-AML conformance.

Document Control

Field	Value
Document	CS-AML Data Model Specification
Version	0.1
Status	Normative baseline
Applies to	CS-AML Framework v0.1 and derivative implementations
Primary audience	Framework maintainers, investigators, data architects, software engineers, security/privacy reviewers, assurance reviewers
Normative terms	SHALL/MUST = mandatory; SHOULD = recommended; MAY = optional
Design objective	Preserve evidentiary provenance, analytical uncertainty, temporal context, and privacy constraints while enabling reusable entity-centric financial intelligence.

1. Purpose and Scope

The purpose of this specification is to define the canonical data structures, semantics, constraints, and lifecycle rules required to represent civil-society financial intelligence under CS-AML. The model is deliberately designed to separate observed information from analytical interpretation and to keep provenance, confidence, and legal/privacy handling attached to the objects they govern.

Core design axiom

Case is context. Entity and Evidence are reusable truth-bearing objects. Analytical conclusions SHALL be represented separately from source material and SHALL remain traceable to the evidence and reasoning that support them.

1.1 In scope

- Case and investigation context
- Source and evidence provenance
- Claim, fact, indicator, and analytical proposition
- Entity identity and entity resolution
- Relationships, ownership, control, and affiliation
- Assets and asset attribution

- Events and temporal state
- Direct, documented, reconstructed, and hypothetical value flows
- Typology matches and indicators
- Hypotheses, intelligence gaps, assessments, and confidence
- Intelligence products, review, dissemination, and closure
- Privacy labels, handling restrictions, audit events, and retention metadata
- Implementation mappings for relational and graph-oriented systems

1.2 Out of scope

- Bank transaction ingestion protocols
- Formal regulated-entity STR/SAR schemas
- Law-enforcement evidentiary rules for criminal prosecution
- Covert collection or unauthorized access mechanisms
- Automated determinations of criminal guilt
- Vendor-specific database design

2. Normative Data Principles

ID	Principle	Normative requirement
DM-P01	Provenance by default	Every substantive evidentiary or analytical object SHALL retain machine-readable provenance to its source, creator, creation time, and applicable verification state.
DM-P02	Separation of observation and inference	Evidence, Fact, Indicator, Hypothesis, and Assessment SHALL be modeled as distinct object classes. Implementations SHALL NOT collapse them into a single notes field.
DM-P03	Temporal truth	Where a statement can change over time, the model SHALL support valid-time boundaries and SHALL avoid treating current state as historically permanent.
DM-P04	Entity reuse	Entities SHALL be reusable across cases. Case membership SHALL be represented as contextual association rather than by duplicating entity records per case.
DM-P05	Identity uncertainty	Entity resolution SHALL support candidate matches, confidence, merge rationale, and reversible decisions.

ID	Principle	Normative requirement
DM-P06	Value-flow epistemics	Direct, documented, reconstructed, and hypothetical flows SHALL be distinguishable in the data model and in any derived visualization.
DM-P07	Privacy as metadata	Sensitivity, access restrictions, retention, and dissemination limits SHALL be attached to data objects and not only to user-interface screens.
DM-P08	Immutability of originals	Original evidence SHALL be preserved. Corrections and annotations SHALL be additive or versioned; they SHALL NOT silently rewrite the historical record.
DM-P09	No guilt flag	The canonical model SHALL NOT contain a boolean field that declares a person guilty of money laundering or other crime.
DM-P10	Auditability	Material changes to identity, evidence, relationships, assessment, or dissemination state SHALL generate auditable events.

3. Canonical Information Architecture

CS-AML defines a layered information model. The layers are semantic rather than deployment-specific; one implementation MAY use PostgreSQL, a graph database, object storage, or a hybrid architecture, provided the normative semantics are preserved.

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CASE CONTEXT
|
+-- SOURCE / EVIDENCE
|   |
|   +-- CLAIM / FACT
|
+-- ENTITY / RELATIONSHIP / ASSET / EVENT
|   |
|   +-- VALUE FLOW
|
+-- INDICATOR / TYPOLOGY MATCH
|   |
|   +-- HYPOTHESIS / INTELLIGENCE GAP
|
+-- ASSESSMENT
|
+-- INTELLIGENCE PRODUCT

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3.1 Canonical object families

Family	Canonical objects
Context	Case, CaseEntity, CaseRole, InvestigationQuestion, ScopeChange
Evidence	Source, EvidenceItem, EvidenceExtract, Claim, Fact
Knowledge graph	Entity, PersonProfile, OrganizationProfile, AccountProfile, Asset, Relationship, Event, Location
Financial analysis	ValueFlow, ValueFlowLeg, OwnershipInterest, ControlAssertion, AssetAttribution
Analytical	Indicator, TypologyMatch, Hypothesis, HypothesisEvidenceLink, IntelligenceGap, Assessment
Product & assurance	IntelligenceProduct, Review, Dissemination, ClosureRecord
Governance	DataClassification, AccessLabel, RetentionRule, AuditEvent, VersionRecord

4. Common Object Envelope

All canonical objects SHALL expose a minimum common envelope. Implementations MAY extend this envelope but SHALL NOT redefine the meaning of the normative fields.

Field	Type	Cardinality	Meaning
id	UUID/string	Required	Globally unique immutable object identifier.
object_type	enum	Required	Canonical class name.
schema_version	string	Required	Schema version under which object was created/validated.
created_at	datetime	Required	System creation timestamp.
created_by	principal-id	Required	Human or service principal responsible for creation.
updated_at	datetime	Required	Last material update

Field	Type	Cardinality	Meaning
			timestamp.
status	enum	Required	Lifecycle status appropriate to object class.
classification	enum	Required	Information sensitivity classification.
access_labels	array	Conditional	Attribute-based handling labels.
case_links	array<case-id>	Optional	Contextual case associations; not ownership.
provenance_refs	array<id>	Conditional	Links to source/evidence/provenance objects.
confidence	object	Conditional	Confidence object where analytical uncertainty exists.
valid_from	datetime/date	Optional	Beginning of real-world validity.
valid_to	datetime/date	Optional	End of real-world validity.
record_version	integer	Required	Monotonic version number for optimistic concurrency/audit.
deleted_at	datetime	Optional	Soft-delete/tombstone marker where policy permits.

5. Identifier and Naming Standard

Object identifiers SHALL be opaque and SHALL NOT encode sensitive real-world identifiers, names, or conclusions. Human-readable display codes MAY be assigned separately.

Object	Prefix example	Canonical ID	Human display code
Case	CASE	UUID	CSAML-CASE-2026-001
Entity	ENT	UUID	ENT-000145
Evidence	EVD	UUID	EVD-000381

Object	Prefix example	Canonical ID	Human display code
Hypothesis	HYP	UUID	HYP-006
Assessment	ASS	UUID	ASS-003

6. Context and Case Objects

6.1 Case

Represents the bounded investigative context in which questions, scope, risks, analytical objects, and products are managed. A Case SHALL NOT own reusable entities or evidence; it SHALL link to them.

Field	Type	Req.	Cardinality	Semantics
title	string	Y	1	Neutral descriptive title.
case_code	string	Y	1	Human-readable identifier.
purpose	text	Y	1	Legitimate investigative purpose.
investigation_questions	ref[]	Y	1..n	Questions to be answered.
jurisdictions	code[]	N	0..n	Relevant jurisdictions.
time_scope	interval	N	0..1	Primary period under review.
lead_analyst	principal	Y	1	Accountable analyst.
risk_rating	enum	Y	1	Operational/harm risk rating.
closure_reason	enum	N	0..1	Reason case closed.

Normative rules:

- Case SHALL contain a documented legitimate purpose before collection begins.
- Case title SHALL avoid presuming criminality.
- Case closure SHALL NOT delete reusable entity or evidence objects.

Lifecycle states:

DRAFT -> AUTHORIZED -> ACTIVE -> REVIEW -> CLOSED -> MONITORING -> REOPENED

6.2 InvestigationQuestion

Represents a testable analytical question that constrains scope and prevents open-ended surveillance.

Field	Type	Req.	Cardinality	Semantics
case_id	ref	Y	1	Parent case context.
question	text	Y	1	Neutral question.
priority	enum	Y	1	Analytical priority.
status	enum	Y	1	Open/answered/ retired.
answer_summary	text	N	0..1	Short evidence- linked answer.

Normative rules:

- Questions SHOULD be framed to permit both incriminating and exculpatory answers.

7. Source, Evidence, Claim, and Fact Model

7.1 Source

Represents the origin or provider of information. A Source describes where information came from; it is not itself the extracted evidentiary proposition.

Field	Type	Req.	Cardinality	Semantics
source_type	enum	Y	1	registry, court_record, media, whistleblower, website, dataset, document, interview, other
publisher_or_origin	string	N	0..1	Originating organization/person .
locator	uri/string	N	0..1	URL, archival location, or repository reference.
accessed_at	datetime	Y	1	Collection/access time.

Field	Type	Req.	Cardinality	Semantics
publication_date	date	N	0..1	When published/issued.
reliability_grade	enum	N	0..1	A-F source reliability.
lawful_access_basis	text/enum	Y	1	Why collection/use is permitted.
archive_ref	ref	N	0..1	Preserved snapshot or evidence item.

Normative rules:

- Source reliability SHALL be separated from credibility of individual information items.
- Anonymous sources SHALL NOT be treated as inherently unreliable; uncertainty SHALL be recorded explicitly.

7.2 EvidenceItem

Represents a preserved evidentiary object such as a file, page image, registry record, transcript, screenshot, export, or structured record.

Field	Type	Req.	Cardinality	Semantics
source_id	ref	Y	1	Origin source.
evidence_type	enum	Y	1	document, image, record, transcript, dataset_row, web_snapshot, other
content_hash	string	Y*	0..1	SHA-256 or equivalent where bytes are available.
storage_ref	uri/ref	Y	1	Controlled storage location.
acquired_at	datetime	Y	1	Acquisition time.
acquired_by	principal	Y	1	Collector.
original_format	string	N	0..1	MIME/type.
verification_status	enum	Y	1	unverified, source-verified, independently-corroborated, disputed

Field	Type	Req.	Cardinality	Semantics
redaction_state	enum	Y	1	none, working_redaction, publication_redaction

Normative rules:

- Original evidence bytes SHALL be immutable after registration except for controlled preservation migrations.
- Derived/redacted copies SHALL be linked to the original by derivation relationship.

Lifecycle states:

COLLECTED -> REGISTERED -> VERIFIED -> SUPERSEDED -> RESTRICTED

7.3 EvidenceExtract

Represents the precise portion of EvidenceItem used analytically.

Field	Type	Req.	Cardinality	Semantics
evidence_id	ref	Y	1	Parent evidence.
locator	string	Y	1	Page, paragraph, timecode, row, bounding region, or equivalent.
extract_text	text	N	0..1	Verbatim/normalized extract when lawful.
extract_hash	string	N	0..1	Integrity value for extracted content.
analyst_note	text	N	0..1	Context note; SHALL remain distinguishable from extract.

7.4 Claim

Represents a proposition asserted by a source or person. A Claim is not automatically accepted as fact.

Field	Type	Req.	Cardinality	Semantics
subject_refs	ref[]	Y	1..n	Entities/events referenced.
predicate	string	Y	1	Controlled or human-readable

Field	Type	Req.	Cardinality	Semantics
				predicate.
object_value	variant	Y	1	Claimed value/entity.
evidence_extract_refs	ref[]	Y	1..n	Supporting extracts.
claimant	ref/string	N	0..1	Who makes the assertion.
credibility_grade	enum	N	0..1	1-6 information credibility.
disputed	boolean	Y	1	Whether contested.

Normative rules:

- Claims SHALL preserve attribution.
- A source assertion SHALL NOT be promoted to Fact solely because it appears in an official document if the document merely records a third-party allegation.

7.5 Fact

Represents a proposition accepted by the investigation as established to the stated confidence threshold.

Field	Type	Req.	Cardinality	Semantics
proposition	text/structured	Y	1	Established proposition.
supporting_evidence	ref[]	Y	1..n	Evidence supporting acceptance.
contradicting_evidence	ref[]	N	0..n	Known contradictory material.
fact_status	enum	Y	1	provisional, established, superseded, disputed
valid_time	interval	N	0..1	When proposition is true in real world.
confidence	confidence	Y	1	Analytical confidence.

Normative rules:

- Fact status SHALL be revisable when materially new evidence emerges.
- Fact objects SHALL NOT encode legal guilt or criminal liability unless directly quoting an authoritative adjudication, in which case attribution SHALL be explicit.

8. Entity and Identity Model

8.1 Entity

Canonical identity-bearing node used across cases.

Field	Type	Req.	Cardinality	Semantics
entity_type	enum	Y	1	person, organization, account, address, domain, phone, email, wallet, property, vehicle, vessel, aircraft, contract, project, other
primary_name	string	Y	1	Preferred display label.
aliases	string[]	N	0..n	Alternative labels.
identifiers	identifier[]	N	0..n	Typed external identifiers.
resolution_status	enum	Y	1	unresolved, resolved, conflicted, merged, split
resolution_confidence	confidence	N	0..1	Identity match confidence.
canonical_parent	ref	N	0..1	Target if merged.
source_refs	ref[]	Y	1..n	Sources establishing identity.

Normative rules:

- Names SHALL NOT be treated as unique identifiers.
- A merged entity SHALL retain links to precursor records and merge rationale.
- Implementations SHALL support reversing a mistaken merge without losing history.

8.2 PersonProfile

Extension of Entity for natural persons. Sensitive attributes SHALL only be stored when relevant and lawful.

Field	Type	Req.	Cardinality	Semantics
entity_id	ref	Y	1	Base Entity.
birth_date	date/partial	N	0..1	Known/partial DOB.
nationalities	code[]	N	0..n	Known nationalities.
occupation	string[]	N	0..n	Relevant occupation/role.
public_official_statuses	enum	N	0..1	none, current, former, unknown; not equivalent to suspicion.
sensitive_attribute_notes	restricted text	N	0..1	Only if strictly necessary and lawful.

Normative rules:

- Protected characteristics SHALL NOT be used as AML indicators by themselves.

8.3 OrganizationProfile

Extension for companies, NGOs, agencies, trusts, partnerships, and other organized bodies.

Field	Type	Req.	Cardinality	Semantics
entity_id	ref	Y	1	Base Entity.
organization_type	enum	Y	1	company, ngo, government, trust, partnership, association, other
registration_number	string	N	0..1	Legal registration identifier.
registration_jurisdiction	code	N	0..1	Jurisdiction.
incorporation_date	date	N	0..1	Formation date.
dissolution_date	date	N	0..1	If dissolved.
registered_address	ref	N	0..1	Address entity.

9. Relationship, Ownership, and Control Model

9.1 Relationship

Represents a typed, evidence-linked edge between entities. Relationships are first-class objects because their source, time, and confidence may differ.

Field	Type	Req.	Cardinality	Semantics
from_entity	ref	Y	1	Origin node.
relationship_type	enum	Y	1	Typed semantic relationship.
to_entity	ref	Y	1	Target node.
directionality	enum	Y	1	directed/symmetric.
valid_from	date/datetime	N	0..1	Relationship start.
valid_to	date/datetime	N	0..1	Relationship end.
supporting_evidence	ref[]	Y	1..n	Evidence.
confidence	confidence	Y	1	Confidence.
relationship_status	enum	Y	1	asserted, established, disputed, superseded

Normative rules:

- Edges in a graph visualization SHALL retain their evidence links in the underlying model.
- A relationship based only on co-occurrence SHALL NOT be mislabeled as control, ownership, or financial transfer.

9.2 OwnershipInterest

Specialized relationship for legal or beneficial ownership.

Field	Type	Req.	Cardinality	Semantics
owner_entity	ref	Y	1	Owner/beneficial owner.
owned_entity_or_asset	ref	Y	1	Company/share/asset.
ownership_type	enum	Y	1	legal, beneficial, economic_interest, nominee_asserted

Field	Type	Req.	Cardinality	Semantics
percentage	decimal	N	0..1	0-100 if known.
basis	text/enum	Y	1	Registry, filing, court record, reconstruction, etc.
supporting_evidence	ref[]	Y	1..n	Evidence.
confidence	confidence	Y	1	Confidence.

Normative rules:

- Legal ownership and beneficial ownership SHALL be stored as distinct types.
- Nominee or beneficial ownership inferred from public-source patterns SHALL be marked as inferred/asserted, not established, unless evidence supports establishment.

9.3 ControlAssertion

Represents control that may exist without formal ownership.

Field	Type	Req.	Cardinality	Semantics
controller	ref	Y	1	Person/entity exercising control.
controlled_entity	ref	Y	1	Target.
control_basis	enum[]	Y	1..n	voting, appointment, financing, contractual, operational, family_proxy, other
control_level	enum	N	0..1	minor, significant, dominant, unknown
supporting_evidence	ref[]	Y	1..n	Evidence.
confidence	confidence	Y	1	Confidence.

10. Asset and Event Model

10.1 Asset

Represents an item or right with economic value.

Field	Type	Req.	Cardinality	Semantics
asset_type	enum	Y	1	property, vehicle, vessel, aircraft, security, company_share, crypto_asset, precious_metal, luxury_good, intellectual_property, other
description	text	Y	1	Neutral description.
legal_owner	ref	N	0..1	Known legal owner.
beneficial_owner_assertions	ref[]	N	0..n	OwnershipInterest refs.
controller_refs	ref[]	N	0..n	Control assertions.
valuation	money/range	N	0..1	Value or range.
valuation_date	date	N	0..1	Valuation date.
valuation_basis	enum/text	N	0..1	Registry, market estimate, appraisal, reported value.
location_ref	ref	N	0..1	Location entity.

Normative rules:

- Observed use or association SHALL NOT be represented as ownership without an ownership basis.

10.2 Event

Represents a temporally bounded occurrence involving one or more entities.

Field	Type	Req.	Cardinality	Semantics
event_type	enum	Y	1	incorporation, appointment, resignation, contract_award, acquisition, disposal, transfer, court_event, payment_event, publication, other
start_time	datetime/date	Y	1	Start/occurrence.

Field	Type	Req.	Cardinality	Semantics
end_time	datetime/date	N	0..1	End if interval.
participant_refs	ref[]	Y	1..n	Entities participating.
location_ref	ref	N	0..1	Location.
evidence_refs	ref[]	Y	1..n	Supporting evidence.
confidence	confidence	Y	1	Confidence.

11. Value-Flow Model

Mandatory distinction

Every ValueFlow SHALL carry an epistemic class: DIRECT, DOCUMENTED, RECONSTRUCTED, or HYPOTHETICAL. Systems SHALL preserve this distinction in storage, APIs, exports, and visualizations.

11.1 ValueFlow

Represents movement, conversion, allocation, or inferred transfer of economic value.

Field	Type	Req.	Cardinality	Semantics
flow_class	enum	Y	1	direct, documented, reconstructed, hypothetical
flow_type	enum	Y	1	payment, contract, subcontract, loan, investment, asset_purchase, asset_sale, grant, donation, dividend, crypto_transfer, value_conversion, other
origin_ref	ref	Y	1	Origin entity/asset/event.
destination_ref	ref	Y	1	Destination entity/asset/event.
amount	money/range	N	0..1	Known/estimated

Field	Type	Req.	Cardinality	Semantics
				amount.
currency	code	N	0..1	Currency if monetary.
occurred_at	datetime/date/interval	N	0..1	Temporal scope.
evidence_refs	ref[]	Y*	0..n	Required for direct/documented/reconstructed.
reconstruction_basis	text	Y*	0..1	Required for reconstructed/hypothetical.
confidence	confidence	Y	1	Confidence.
disclaimer	text	N	0..1	Required presentation caveat if needed.

Normative rules:

- DIRECT flows require evidence of the actual movement of value.
- DOCUMENTED flows require an authoritative or transactional document evidencing a transfer/obligation but not necessarily settlement.
- RECONSTRUCTED flows SHALL identify the inferential steps linking origin and destination.
- HYPOTHETICAL flows SHALL never be included in an external intelligence product as if observed.

11.2 ValueFlowLeg

Optional decomposition of a complex flow into ordered legs.

Field	Type	Req.	Cardinality	Semantics
value_flow_id	ref	Y	1	Parent flow.
sequence	integer	Y	1	Order.
origin_ref	ref	Y	1	Leg origin.
destination_ref	ref	Y	1	Leg destination.
amount	money/range	N	0..1	Leg amount.
flow_class	enum	Y	1	Epistemic class for this leg.
evidence_refs	ref[]	N	0..n	Evidence for leg.

12. Indicator and Typology Model

12.1 Indicator

Represents an observed condition relevant to analysis. It SHALL NOT be treated as proof of wrongdoing.

Field	Type	Req.	Cardinality	Semantics
indicator_code	string	Y	1	Stable catalogue or local code.
indicator_class	enum	Y	1	mechanism, corroborating, contextual, disconfirming, gap
description	text	Y	1	Observed condition.
subject_refs	ref[]	Y	1..n	Affected entities/flows/events .
evidence_refs	ref[]	Y	1..n	Evidence.
status	enum	Y	1	observed, corroborated, disputed, retired
confidence	confidence	Y	1	Confidence.

12.2 TypologyMatch

Represents analytical consistency between case evidence and a catalogue typology.

Field	Type	Req.	Cardinality	Semantics
typology_id	string/ref	Y	1	Catalogue entry.
subject_refs	ref[]	Y	1..n	Entities/flows under analysis.
indicator_refs	ref[]	Y	1..n	Observed indicators.
disconfirming_refs	ref[]	N	0..n	Contrary evidence/indicators.
consistency_level	enum	Y	1	no_basis, weak, plausible, strong, compelling
analyst_rationale	text	Y	1	Why level was

Field	Type	Req.	Cardinality	Semantics
				assigned.
confidence	confidence	Y	1	Confidence.

Normative rules:

- TypologyMatch SHALL express consistency, not guilt or legal classification.
- Strong or compelling consistency SHOULD require multiple independent indicators or direct authoritative evidence, consistent with the Typology Catalogue.

13. Hypothesis, Gap, and Assessment Model

13.1 Hypothesis

Represents a testable analytical explanation.

Field	Type	Req.	Cardinality	Semantics
statement	text	Y	1	Specific falsifiable hypothesis.
hypothesis_type	enum	N	0..1	ownership, control, value_flow, typology, source_of_funds, other
supporting_refs	ref[]	N	0..n	Evidence/facts/indicators supporting.
contradicting_refs	ref[]	N	0..n	Evidence/facts contradicting.
alternative_hypothesis_refs	ref[]	N	0..n	Competing explanations.
status	enum	Y	1	open, supported, weakened, rejected, inconclusive
confidence	confidence	Y	1	Current confidence.
next_test	text	N	0..1	Most useful discriminating test.

Normative rules:

- A hypothesis SHALL remain open to disconfirmation.
- Rejected hypotheses SHOULD remain auditable rather than deleted.

13.2 IntelligenceGap

Represents a material unknown that limits assessment.

Field	Type	Req.	Cardinality	Semantics
description	text	Y	1	Unknown or missing information.
importance	enum	Y	1	low, medium, high, critical
related_refs	ref[]	Y	1..n	Hypotheses/questions affected.
collection_feasibility	enum	Y	1	available, difficult, unavailable, unlawful, out_of_scope
status	enum	Y	1	open, partially_resolved, resolved, accepted

13.3 Assessment

Represents a reasoned analytical judgment supported by evidence and explicit confidence.

Field	Type	Req.	Cardinality	Semantics
judgment	text	Y	1	Analytical conclusion.
scope_refs	ref[]	Y	1..n	Question/hypothesis/subjects addressed.
supporting_refs	ref[]	Y	1..n	Facts/indicators/evidence.
limitations	text	Y	1	Known constraints and gaps.
alternative_explanations	text/ref[]	Y	1..n	Material alternatives.
confidence	confidence	Y	1	Confidence in judgment.
review_status	enum	Y	1	draft, peer_reviewed, approved, superseded

Normative rules:

- Assessment language SHALL distinguish known, assessed, and unknown information.
- Assessment SHALL NOT imply criminal guilt beyond the available evidence and mandate.

14. Confidence and Source Evaluation Model

CS-AML separates source reliability, information credibility, and analyst confidence.

Implementations SHALL NOT compress these dimensions into a single unexplained score.

Dimension	Allowed values	Meaning
Source reliability	A-F	A highly reliable; B generally reliable; C mixed; D generally unreliable; E unreliable; F unknown
Information credibility	1-6	1 independently confirmed; 2 probably true; 3 possibly true; 4 doubtful; 5 improbable; 6 cannot be judged
Analyst confidence	Low / Moderate / High	Overall confidence in analytical judgment, based on evidence quality, independence, consistency, and remaining gaps

14.1 Confidence

Structured confidence object embedded/referenced by analytical objects.

Field	Type	Req.	Cardinality	Semantics
level	enum	Y	1	low, moderate, high
basis	text	Y	1	Reason for confidence.
source_diversity	enum	N	0..1	single, multiple_related, multiple_independent
material_gaps	ref[]	N	0..n	Gap references.
last_reviewed_at	datetime	Y	1	Review timestamp.
reviewer	principal	N	0..1	Reviewer if required.

15. Intelligence Product and Dissemination Model

15.1 IntelligenceProduct

Represents an approved analytical output for an identified audience.

Field	Type	Req.	Cardinality	Semantics
product_type	enum	Y	1	financial_intelligence_note, investigation_brief, entity_profile, asset_profile, network_analysis, referral_package, publication_support
title	string	Y	1	Neutral title.
assessment_refs	ref[]	Y	1..n	Underlying assessments.
evidence_manifest	ref[]	Y	1..n	Evidence inventory.
audience	enum/string[]	Y	1..n	Authorized audience.
classification	enum	Y	1	Sensitivity.
approval_state	enum	Y	1	draft, reviewed, approved, disseminated, withdrawn
version	string	Y	1	Product version.
published_at	datetime	N	0..1	If published/disseminated.

15.2 Review

Represents peer, legal, privacy, security, or red-team review.

Field	Type	Req.	Cardinality	Semantics
review_type	enum	Y	1	peer, legal, privacy, security, red_team, editorial
target_ref	ref	Y	1	Object/product reviewed.
reviewer	principal	Y	1	Reviewer.

Field	Type	Req.	Cardinality	Semantics
findings	text/structured	Y	1	Review findings.
decision	enum	Y	1	approve, approve_with_changes, return, reject
completed_at	datetime	Y	1	Completion time.

15.3 Dissemination

Records controlled release of a product or data package.

Field	Type	Req.	Cardinality	Semantics
product_ref	ref	Y	1	Product disseminated.
recipient	string/ref	Y	1	Recipient/authority/audience.
purpose	text	Y	1	Purpose.
released_at	datetime	Y	1	Release time.
released_by	principal	Y	1	Authorizer.
handling_caveats	string[]	N	0..n	Restrictions.
receipt_or_reference	string	N	0..1	Acknowledgement/reference.

16. Privacy, Classification, and Access-Control Metadata

Classification	Meaning
PUBLIC	Suitable for public release after normal review.
INTERNAL	Routine internal operational information.
SENSITIVE	Could create privacy, reputational, safety, or investigative harm if disclosed.
RESTRICTED	High-risk information requiring named-role or case-specific authorization.
SOURCE-PROTECTED	Information whose disclosure could identify or endanger a confidential source.

Access labels MAY add purpose, jurisdiction, source-protection, embargo, legal-review, or compartment restrictions. Implementations SHALL enforce the most restrictive applicable label.

16.1 RetentionRule

Represents retention and disposition policy attached to objects or classes.

Field	Type	Req.	Cardinality	Semantics
rule_id	string	Y	1	Policy identifier.
applies_to	enum/ref	Y	1	Object type/object.
retention_period	duration/date	Y	1	Retention term.
trigger	enum	Y	1	creation, case_closure, dissemination, legal_hold_release, other
disposition	enum	Y	1	delete, anonymize, archive, review
legal_hold	boolean	Y	1	Whether disposition suspended.

17. Temporal, Versioning, and Audit Semantics

CS-AML distinguishes system time (when the system knew/recorded something) from valid time (when the represented fact/relationship was true in the real world). Where historically material, both SHOULD be stored.

valid_from / valid_to = real-world validity
created_at / updated_at = system record time
record_version = object mutation sequence
AuditEvent = who changed what, when, and why

17.1 AuditEvent

Append-only record of material actions.

Field	Type	Req.	Cardinality	Semantics
actor	principal	Y	1	Human/service principal.
action	enum/string	Y	1	create, update, merge, split, review, approve, disseminate, restrict, delete,

Field	Type	Req.	Cardinality	Semantics
				export, login_sensitive, other
target_ref	ref	Y	1	Affected object.
occurred_at	datetime	Y	1	Time.
reason	text	N	0..1	Reason/context.
before_hash	string	N	0..1	Optional integrity reference.
after_hash	string	N	0..1	Optional integrity reference.
case_context	ref	N	0..1	Case context if applicable.

Normative rules:

- Audit events SHALL be append-only for normal users.
- Administrative correction of audit data SHALL itself generate an audit event.

18. Cross-Object Integrity Rules

Rule	Constraint
DM-I01	Every Fact SHALL reference at least one EvidenceItem or EvidenceExtract.
DM-I02	Every Indicator SHALL reference supporting evidence and at least one subject object.
DM-I03	Every TypologyMatch SHALL reference one catalogue typology and one or more indicators.
DM-I04	Every Assessment SHALL reference supporting analytical objects and include limitations.
DM-I05	A ValueFlow with flow_class=DIRECT SHALL NOT exist without direct evidence of movement.
DM-I06	A Relationship of type OWNS/BENEFICIAL_OWNER_OF/CONTROLS SHALL include basis and confidence.
DM-I07	Entity merge SHALL preserve precursor identifiers and generate AuditEvent.
DM-I08	External dissemination SHALL reference an approved

Rule	Constraint
	IntelligenceProduct.
DM-I09	Source-protected information SHALL NOT be exported into lower-classification products without explicit de-identification review.
DM-I10	Deletion/anonymization SHALL respect evidence preservation obligations and legal holds.

19. Canonical Graph Mapping

Graph implementations SHOULD use Entity, Asset, Event, and selected contextual objects as nodes, with Relationship and ValueFlow as first-class edges or edge-objects. Evidence SHALL NOT be lost when graph projections are materialized.

```
(:Person)-[:DIRECTOR_OF {relationship_id,...}]->(:Organization)
(:Person)-[:BENEFICIAL_OWNER_OF {ownership_interest_id,...}]->(:Organization)
(:Organization)-[:AWARDED_CONTRACT]->(:Contract)
(:Organization)-[:ACQUIRED]->(:Asset)
(:Entity)-[:VALUE_FLOW {value_flow_id, flow_class,...}]->(:Entity)
```

19.1 Graph projection rule

A graph edge is a projection of a canonical Relationship or ValueFlow object. The canonical object remains authoritative because it contains provenance, validity period, confidence, and handling metadata that may exceed native graph-edge capabilities.

20. Reference Relational Mapping

```
cases
sources
evidence_items
evidence_extracts
claims
facts
entities
entity_identifiers
relationships
ownership_interests
control_assertions
assets
events
value_flows
value_flow_legs
indicators
typology_matches
hypotheses
```

hypothesis_links
intelligence_gaps
assessments
intelligence_products
reviews
disseminations
audit_events

This list is illustrative. Implementations MAY normalize or denormalize differently provided semantic and integrity requirements are preserved.

21. Canonical JSON Example

```
{
  "id": "vf-7c2e...",
  "object_type": "ValueFlow",
  "schema_version": "0.1",
  "flow_class": "reconstructed",
  "flow_type": "asset_purchase",
  "origin_ref": "org-123",
  "destination_ref": "asset-456",
  "amount": {
    "value": 8500000000,
    "currency": "IDR",
    "precision": "estimated"
  },
  "occurred_at": {
    "from": "2026-05-01",
    "to": "2026-05-31"
  },
  "evidence_refs": [
    "evd-11",
    "evd-19"
  ],
  "reconstruction_basis": "Contract award and subsequent deed records indicate a plausible economic link; no bank settlement record is available.",
  "confidence": {
    "level": "moderate",
    "basis": "Two independent documentary sources; settlement route unknown."
  },
  "classification": "SENSITIVE",
  "record_version": 3
}
```

22. API and Exchange Requirements

- APIs SHALL expose object IDs, schema version, record version, classification, and provenance references.

- APIs SHOULD support field-level filtering to prevent unnecessary disclosure of sensitive attributes.
- Bulk exports SHALL preserve relationship/evidence IDs so provenance survives movement between systems.
- Systems SHALL reject writes that violate mandatory cross-object integrity rules.
- Machine-generated analytical suggestions SHALL be marked with creator_type=automation/AI and SHALL require human review before becoming an Assessment.

23. Interoperability and External Identifiers

The canonical model is implementation-neutral and MAY map to external ontologies or identifiers. External IDs SHALL be stored as typed identifiers and SHALL NOT replace internal canonical IDs.

Domain	Examples	Model treatment
Corporate	registration number, LEI, registry ID	Entity.identifier[]
Persons	official registry identifiers where lawful	Entity.identifier[] with restricted handling where sensitive
Sanctions/PEP	list-specific subject IDs	External identifier + Source/Evidence
Crypto	wallet/address	Entity type=wallet; chain/network as typed attribute
Procurement	tender/contract IDs	Contract Entity/Event/Source refs
Geospatial	parcel/registry identifiers	Asset/Location identifiers

24. Schema Versioning and Migration

- Every stored object SHALL declare schema_version.
- Breaking semantic changes SHALL increment the major/minor model version according to framework governance.
- Migrations SHALL preserve provenance and audit history.
- A migration SHALL NOT silently convert reconstructed value flow into direct value flow, or claim into fact, or hypothesis into assessment.
- Deprecated fields SHOULD remain readable for a defined transition period.

25. Data Quality and Validation

Quality dimension	Expected measure
Completeness	Required fields populated and required links present.

Quality dimension	Expected measure
Provenance coverage	Share of facts/indicators/assessments traceable to evidence.
Entity resolution quality	Merge/split error rate; unresolved duplicate rate.
Temporal quality	Coverage of validity dates for time-sensitive relationships.
Confidence hygiene	Analytical objects with explicit confidence basis.
Classification hygiene	Sensitive objects carrying appropriate handling labels.
Orphan rate	Objects lacking expected parent/context/evidence links.
Audit completeness	Material actions represented by AuditEvent.

26. Data Model Conformance

An implementation claiming CS-AML Data Model v0.1 conformance SHALL demonstrate the following minimum capabilities:

- Persistent canonical IDs and schema versioning.
- Separate Source, Evidence, Fact, Indicator, Hypothesis, Assessment, and IntelligenceProduct objects.
- Reusable Entity objects with evidence-based resolution history.
- Evidence-linked Relationship and ValueFlow representations.
- Explicit value-flow epistemic class.
- Confidence and intelligence-gap representation.
- Sensitivity/classification metadata and access enforcement.
- Append-only audit history for material actions.
- Export that preserves provenance references and object identity.
- Validation of the mandatory cross-object integrity rules in Section 18.

Conformance principle

A schema is not conformant merely because it uses similar table or class names. Conformance depends on preserving the semantics, separations, provenance requirements, and integrity constraints defined by this specification.

Annex A. Controlled Enumerations (Baseline)

Enumeration	Baseline values
entity_type	person, organization, account, address, domain, phone, email, wallet, property, vehicle, vessel, aircraft, contract, project, other
flow_class	direct, documented, reconstructed, hypothetical
confidence.level	low, moderate, high
classification	PUBLIC, INTERNAL, SENSITIVE, RESTRICTED, SOURCE-PROTECTED
hypothesis.status	open, supported, weakened, rejected, inconclusive
typology_match.consistency_level	no_basis, weak, plausible, strong, compelling
review.decision	approve, approve_with_changes, return, reject

Annex B. Canonical Relationship Vocabulary (Baseline)

Relationship	Direction	Semantics
OWNS	Directed	Legal ownership unless subtype says otherwise.
BENEFICIAL_OWNER_OF	Directed	Beneficial ownership/economic interest.
CONTROLS	Directed	Control without requiring ownership.
DIRECTOR_OF	Directed	Corporate role.
OFFICER_OF	Directed	Organizational role.
EMPLOYED_BY	Directed	Employment/role.
RELATED_TO	Symmetric	Familial/known relation; subtype required where possible.
ASSOCIATE_OF	Symmetric	Documented association; SHALL NOT imply wrongdoing.
SHARES_ADDRESS_WITH	Symmetric	Common address.
SHARES_PHONE_WITH	Symmetric	Common phone.
SHARES_DEVICE_WITH	Symmetric	Common device if lawfully obtained.

Relationship	Direction	Semantics
CONTRACTED_BY	Directed	Contract relationship.
SUPPLIER_TO	Directed	Supply relationship.
ACQUIRED	Directed	Entity acquired asset.
SOLD_TO	Directed	Asset transfer.
REPRESENTED_BY	Directed	Professional/legal representation.
PAID_BY	Directed	Documented payer relationship, not necessarily direct bank proof.

Annex C. Minimal Implementation Object Set

A minimum viable conformant implementation SHOULD support at least the following persistent objects:

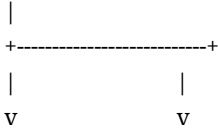
Case
Source
EvidenceItem
EvidenceExtract
Fact
Entity
Relationship
Asset
Event
ValueFlow
Indicator
TypologyMatch
Hypothesis
IntelligenceGap
Assessment
IntelligenceProduct
Review
Dissemination
AuditEvent

Annex D. Example Analytical Chain

SOURCE
Government procurement portal
|
v
EVIDENCE
Contract award record
|
v

FACT

Company A received Contract X on 12 May



ENTITY / RELATIONSHIP EVENT

Company A --DIRECTOR_OF--> Contract award
Person B



INDICATOR

Related entity acquired property shortly after award



HYPOTHESIS

Value may have been diverted through related entity



VALUE FLOW (RECONSTRUCTED)

Contract -> Company A -> related entity -> asset



ASSESSMENT

Pattern is consistent with procurement-value diversion;
settlement route remains unknown.



INTELLIGENCE PRODUCT